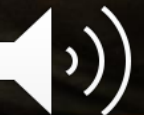


ECON1010

Introductory Microeconomics

Tutorial 10 | Externalities and Property Rights



This Week in ECON1010

CML 5 is due **next week** 4.00pm May 19

Assessment

85% **7**

75% **6**

65% **5**

50% **4**

47% **3**

30% **2**

0% **1**

CMLs (25%)

- Computer Managed Learning
- Each worth 5%
- Best 5/6 count towards final grade

CLEAR-JE (20%)

- Contextualised Learning of Economics via Authentic Reflection using Journal Entries
- Each worth 10%
- Best 2/3 count towards final grade

Final Exam (55%)

Tools and Resources

I need HELP...

1. **Ed Discussion Board**
(Blackboard)
2. **Consultation** (8.00am
Mon, 39-125A)
3. **Staff email**
econ1010@uq.edu.au

I REALLY need HELP...

- <https://my.uq.edu.au/contact/student-central>

ECON1010 Consult Timetable

Semester 1 2026

Time	Mon	Tue	Wed	Thu	Fri
8:00-8:30	CONSULT - SAM 39-125A				
8:30-9:00					
9:00-9:30					
9:30-10:00					
10:00-10:30				CONSULT - LUCAS 39-125A	
10:30-11:00					
11:00-11:30	CONSULT - GABE 39-125A	CONSULT - BORIS 39-125A	CONSULT - IKMAL 39-125A	CONSULT - NIK 39-125A	CONSULT - CARL 39-641
11:30-12:00					
12:00-12:30					
12:30-13:00					
13:00-13:30					LECTURE
13:30-14:00		CONSULT - DYLAN 39-125A			
14:00-14:30				CONSULT - GUANCHU 39-125A	
14:30-15:00					
15:00-15:30					
15:30-16:00					
16:00-16:30			CONSULT - KIRTAN 39-125A	CONSULT - NANDAN 39-125A	
16:30-17:00					
17:00-17:30					

Student Central

Need help? We've got you covered.

Contact us

Monday to Friday

- ✉ Email
- ☎ 1300 275 870 (8.30am-5pm)
- 📍 Building 42, St Lucia
- 📍 Building 8101A, Gatton

Chat – 1:51 (1 Minute) wait

Emergency help

For immediate risk:

- UQ Campus Security
- ☎ 07 3365 3333 (24/7)

Off-campus emergency

- ☎ 000 (24/7)

Crisis support

For urgent mental health support:

- UQ Counselling and Crisis Line
- ☎ 1300 851 998 (24/7)

Text a Crisis Counsellor

- ☎ 0488 884 115 (4.30pm-8am)

Sexual Misconduct Support Unit (SMSU)

Externalities and Property Rights

Question 1a

In your own words what are:

- i) Negative externalities (list one example)
- ii) Positive externalities (list one example)
- iii) Different ways to deal with externalities (list two methods)



Question 1a

ii) Negative externalities (list one example)

Costs of an activity for those other than the individual consumer or supplier. Negative externalities are shown in the Marginal Cost(Social) curve.

Example: Smoking has negative externalities when consumed. Firstly, second hand smoking as it affects the health of those around the consumer. Secondly, the medical costs paid by government (with taxes) as smokers fall ill to go to hospital.

ii) Positive externalities (list one example)

Benefits of an activity received by those other than the individual consumer or supplier. Positive externalities are shown in the Marginal Benefit(Social) curve.

Example: Flu vaccinations have the private benefit of preventing the individual from getting sick, but also have the positive externality of reducing the spread of infection to others. There is also a more general benefit to society as the individual is less likely to take sick days and miss work or important university lectures.

Question 1a

ii) Different ways to deal with externalities (list two methods)

1. **Private remedies** – Agreement between parties to reduce or promote the externality for a more beneficial outcome for the parties involved. Coase Theorem is a theory based on this ‘Private Remedy’. No matter how property (or legal) rights are distributed amongst parties, as long as there is no costs in negotiating, the parties will naturally reach agreement for a socially efficient outcome.
2. **Command and Control** – Government creating laws to enforce, reduce or ban certain actions that produce externalities on society. Failure to comply with the Command and Control usually results in a form of punishment (financial or criminal penalty)
3. **Taxes and subsidies** – The use of financial incentives to affect the behaviour of the market. That is taxes to discourage and subsidies to encourage.
4. **Tradable permits** – The issuing of ‘permits’ to operate with negative externalities (often pollution) to achieve the most efficient and mutually beneficial outcome. The permits are issued (either for free or sold) to firms and then traded amongst firms.

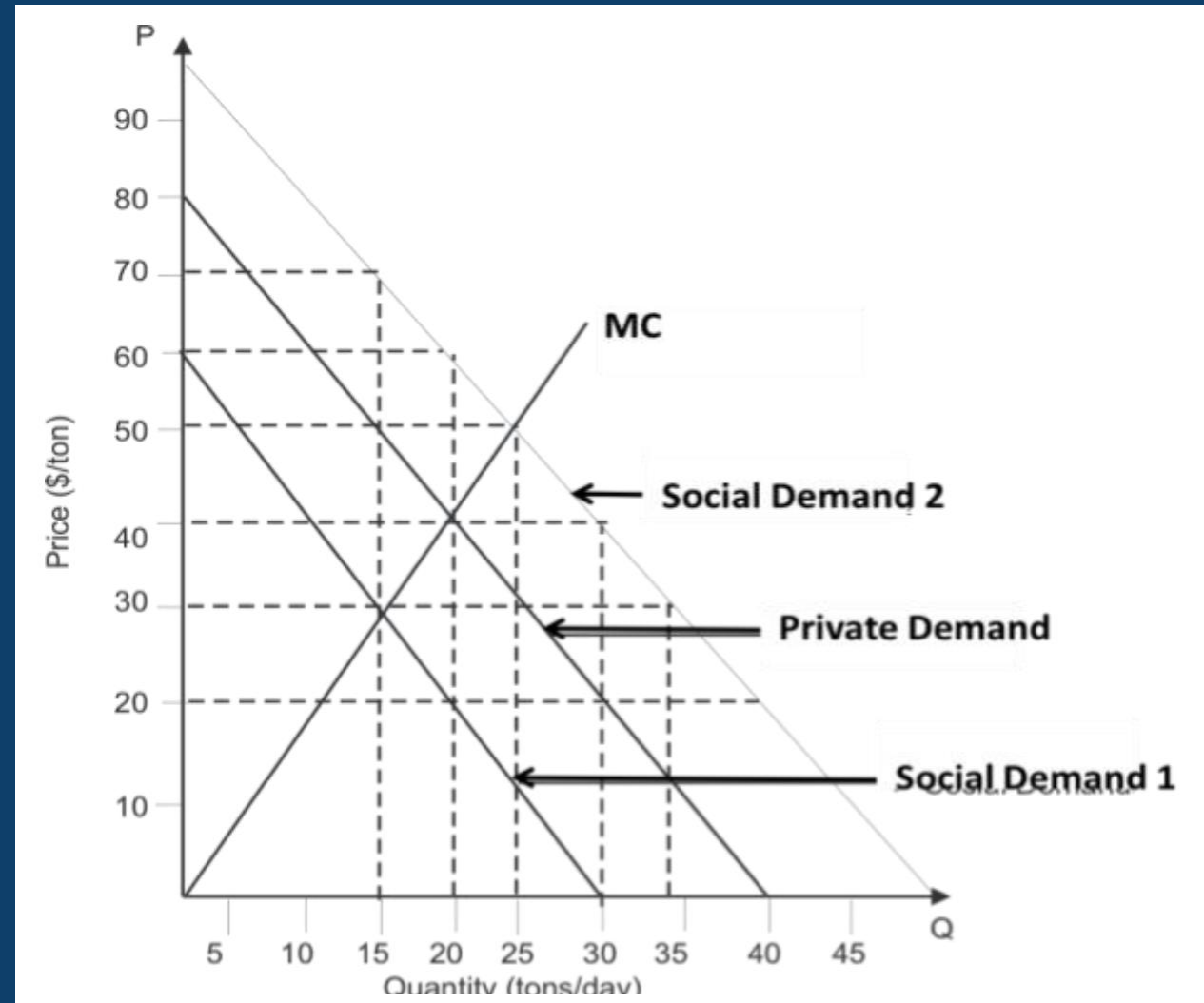
The rationale is that firms producing many negative externalities are without (or have low) profits and so cannot afford the permits. These low-profit, high polluters therefore must shut down or switch to alternative production methods. Higher-profit companies with low negative externalities can purchase these limited permits from their competitors. This reduces overall negative externalities (to the number of permits allocated) while minimising the financial cost to the market.

Question 1b

Consider the following diagram.

Suppose that production of this good is accompanied by an external benefit. The private market equilibrium quantity is _____ the socially optimal quantity.

- a. equal to
- b. 10 tons/day less than
- c. 10 tons/day more than
- d. 5 tons/day more than
- e. 5 tons/day less than

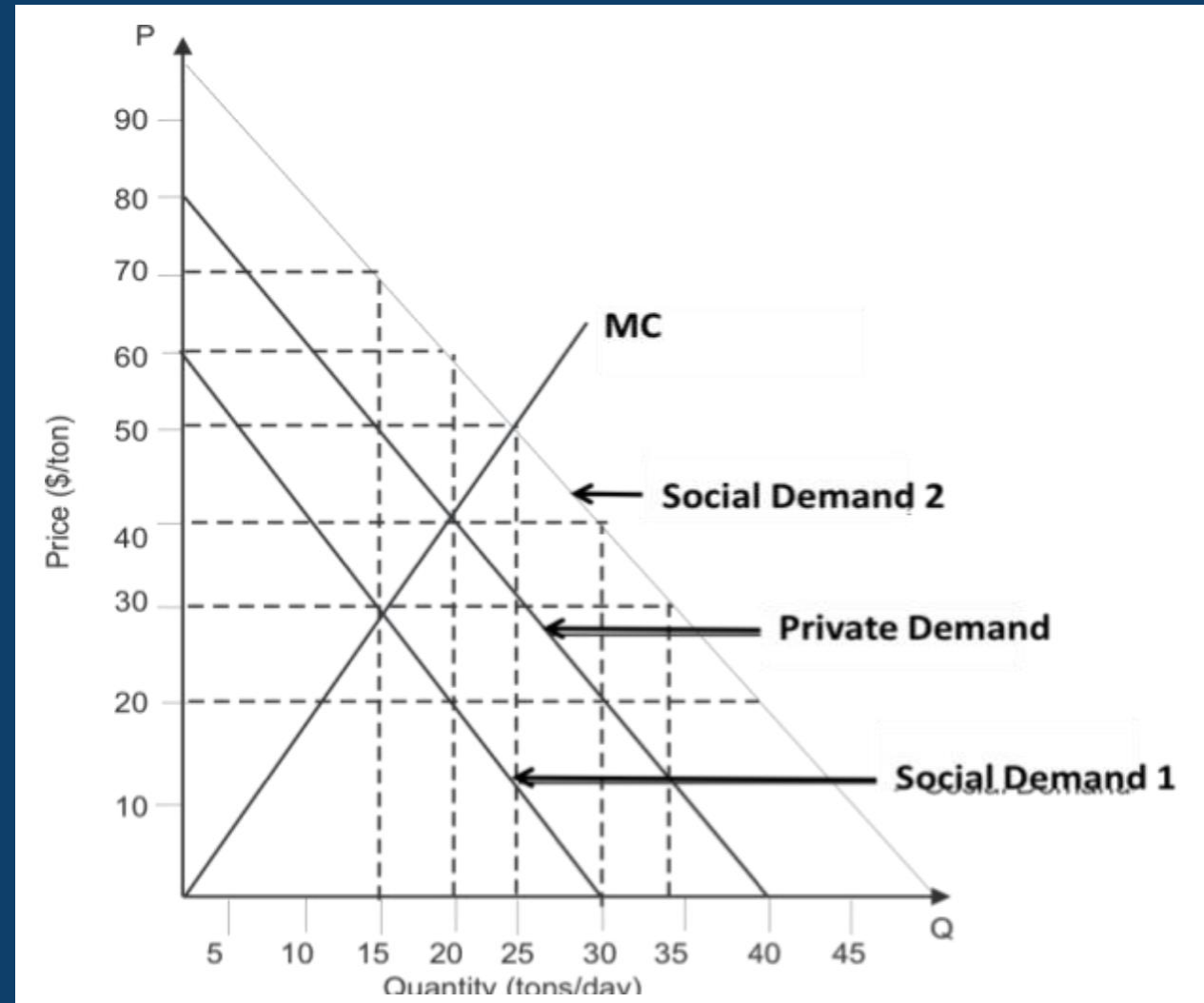


Question 1b

Consider the following diagram.

Suppose that production of this good is accompanied by an external benefit. The private market equilibrium quantity is _____ the socially optimal quantity.

- a. equal to
- b. 10 tons/day less than
- c. 10 tons/day more than
- d. 5 tons/day more than
- e. 5 tons/day less than

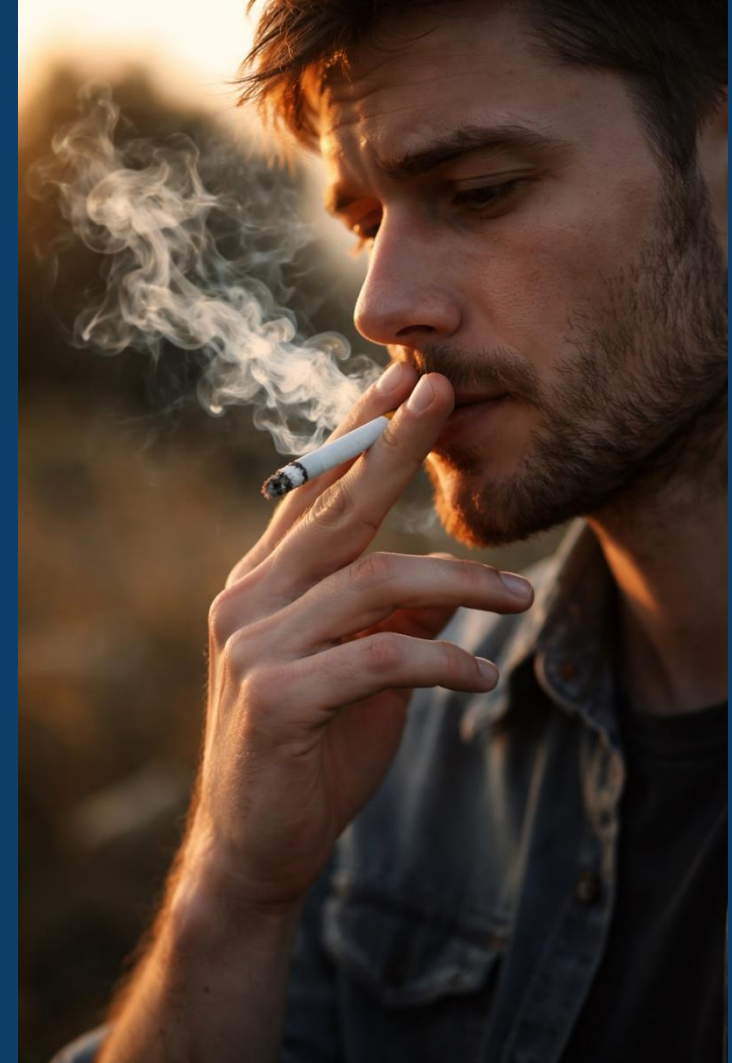


Social Demand with an external benefit will be greater than private demand, therefore Social Demand 2 is the correct curve.

Question 2a

Consider the impact of smoking cigarettes in society.

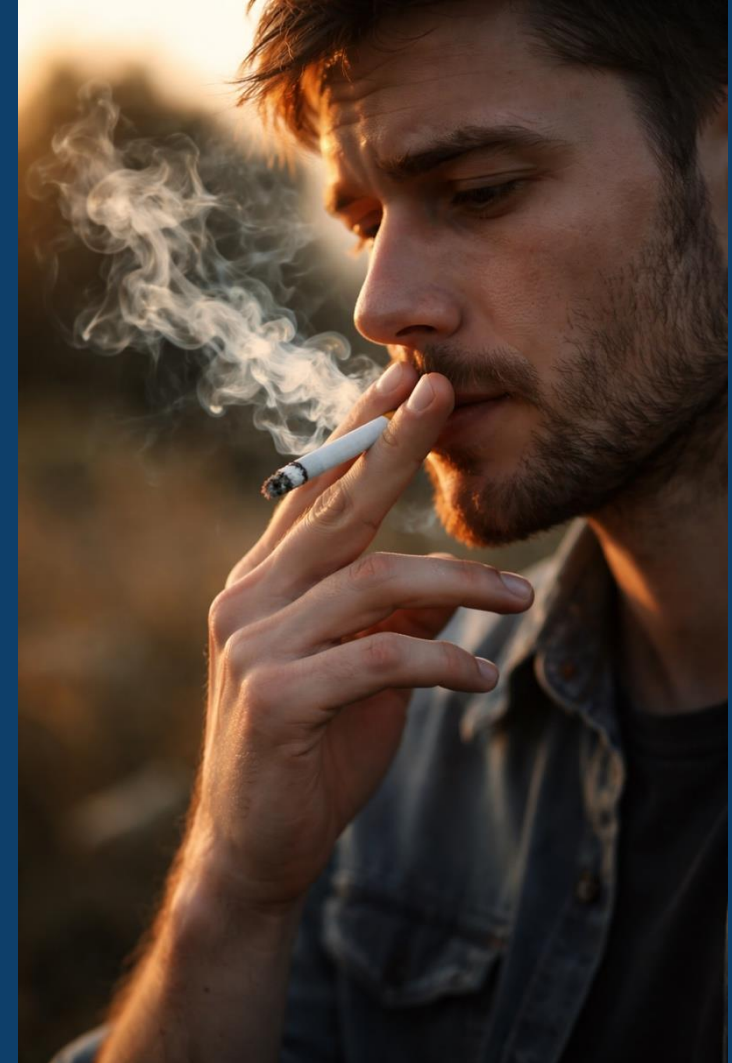
- i) What negative externalities can come from smoking?
- ii) Draw the market graph and show the externalities in an additional curve.
- iii) According to your graph above, should quantity increase or decrease for the most efficient social outcome?
- iv) Shade in the deadweight loss.



Question 2a

i) What negative externalities can come from smoking?

1. **Cost to the public health care system**
2. **Second hand smoking**
3. **Pollution when cigarette butts are left on the street**
4. **Air pollution**
5. **Pesticides in growing the tobacco crop**
6. **Carbon dioxide emissions in the production and transportation of cigarettes**
7. **Poor work conditions for labourers in developing countries**



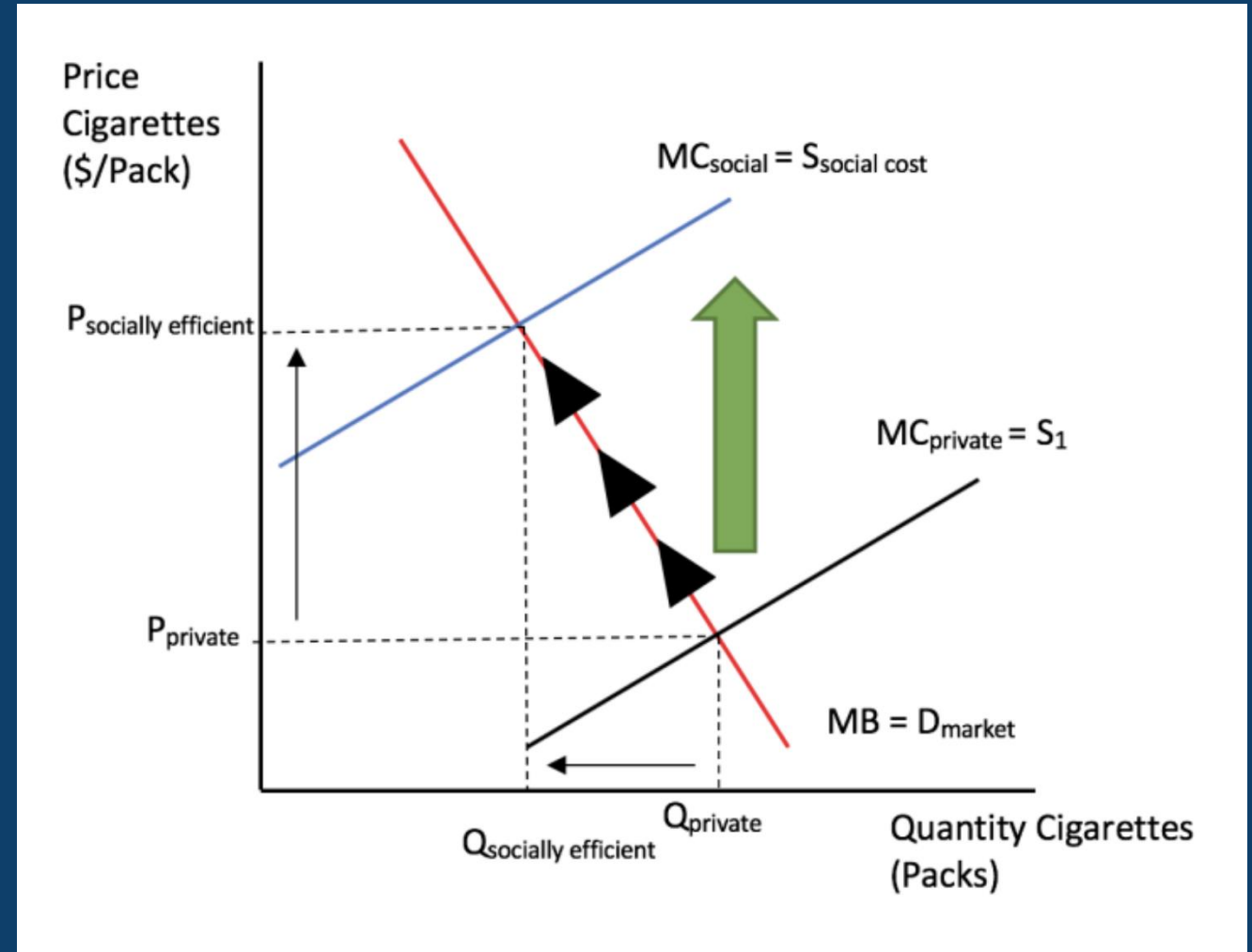
Question 2a

ii) Draw the market graph and show the externalities in an additional curve.

As smoking causes negative externalities, the extra costs will be shown in the Marginal Cost_(Social) curve.

Positive = $MSB = MPB + \text{External Benefit}$

Negative = $MSC = MPC + \text{External Benefit}$

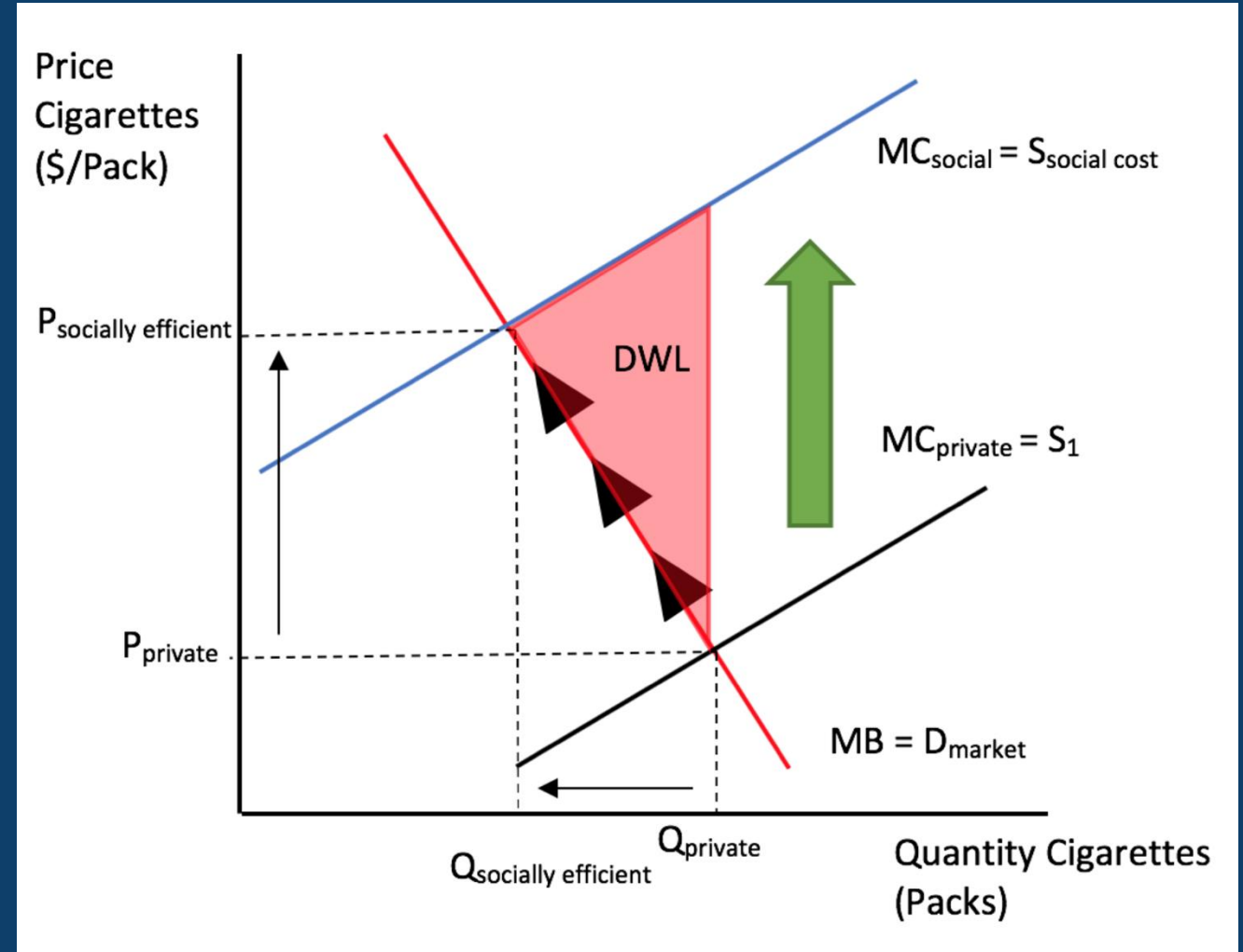


Question 2a

iii) According to your graph above, should quantity increase or decrease for the most efficient social outcome?

Quantity should decrease from Q_{private} down to $Q_{\text{socially efficient}}$.

iv) Shade in the deadweight loss.



Question 2b

A company called Tough Textiles is considering disposing its chemical waste “for free” by dumping it into a nearby river. However, harmful toxins in the waste will affect the river’s water quality, reducing the production of fish at a downstream fish farm and number of chickens on a chicken farm. At a cost, Tough Textiles can filter out the harmful toxins in the chemical waste before dumping it into the river. If this is done, there will be no effect on existing levels of fish and chicken production. The fish and chicken farmers will therefore suffer no damage to their businesses.

The relevant net gains (in thousands of dollars) for the three parties involved, depending on if the filter is used or not, are listed below.

	With Filter ('000s)	Without Filter ('000s)
Tough Textiles	\$200	\$400
Fish Farmer	\$180	\$50
Chicken Farmer	\$130	\$25
Net Community Gain		

i) If Tough Textiles does not install the filter, the dumping of waste will cause a net social loss/gain of how many thousands of dollars? (Hint: compare net gains in the community with and without the filter)

ii) The local fish farmer and chicken farmer wish to deal with the externality and are considering paying Tough Textiles to install a filter. Fill out the table below. How much benefit do the fish and chicken farmer receive with the filter installed:

	With Filter ('000s)	Without Filter ('000s)	Benefit of Filter ('000s)
Tough Textiles	\$200	\$400	-\$200
Fish Farmer	\$180	\$50	
Chicken Farmer	\$130	\$25	

iii) How much will the fish and chicken farmer be willing to pay collectively for Tough Textiles to install a filter?

iv) If all three parties can communicate with each other at no cost, will Tough Textiles use a filter (ie. can the farmers pay enough to compensate Tough Textiles’ loss)?

v) Consider that the fisherman and the chicken farmer now have a legal right to a clean river. Would Tough Textiles use a filter?



Question 2b

- i) If Tough Textiles does not install the filter, the dumping of waste will cause a net social loss/gain of how many thousands of dollars? (Hint: compare net gains in the community with and without the filter)
- Currently without the filter the market is producing \$475,000 in net gains. With the filter the market would produce \$510,000 in net gains. Therefore, while Tough Textiles doesn't filter their water there is a net social loss of \$35,000.**
- ii) The local fish farmer and chicken farmer wish to deal with the externality and are considering paying Tough Textiles to install a filter. Fill out the table below. How much benefit do the fish and chicken farmer receive with the filter installed:

	With Filter (‘000s)	Without Filter (‘000s)	Benefit of Filter (‘000s)
Tough Textiles	\$200	\$400	-\$200
Fish Farmer	\$180	\$50	\$130
Chicken Farmer	\$130	\$25	\$105

The fish farm receives an additional benefit of \$130,000 with the filter installation. The chicken farmer receives an additional benefit of \$105,000 with the filter installation.

Question 2b

iii) How much will the fish and chicken farmer be willing to pay collectively for Tough Textiles to install a filter?

The fish and chicken farm will be willing to pay up to \$235,000 to install the filter.

iv) If all three parties can communicate with each other at no cost, will Tough Textiles use a filter (ie can the farmers pay enough to compensate for Tough Textiles' loss)?

Yes, the fish and chicken farmer can pay \$200,001 to Tough Textiles to convince the company to install the filter.

v) Consider that the fisherman and the chicken farmer now have a legal right to a clean river. If someone dirties the river, the farmers can take this person to court to recover all of their financial losses. Would Tough Textiles use a filter?

This is the crux of the Coase Theorem – when property rights are assigned and transaction costs are zero (or at least very low) then private negotiations will gravitate towards socially efficient and mutually beneficial outcomes. In this case Tough Textiles would install a filter because this is cheaper (\$200) than paying the fisherman's losses ($130+105=\$235$).

Question 3

Consider the following to develop your understanding of how externalities and economics apply in daily life.

- i) Consider a negative externality that affects you on a daily basis.
- ii) How does this negative externality affect you?
- iii) Which of the four methods of dealing with externalities are available to solve this negative externality?



Thank You

Next week: Public Goods

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